

# SHICHENG LIU

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## EDUCATION

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Stanford University

**Ph.D. in Computer Science**

2026 (*expected*)

Natural Language Processing Group, 4th year

2025-2026 IBM PhD Fellowship Recipient

The University of Chicago

**(Honors) B.S. Computer Science** *with a specialization in Computer Systems*

Jun. 2022

**(Honors) B.S. Mathematics**

**Minor in Physics**

Cumulative GPA: 3.985/4.000 (*summa cum laude*)

California Institute of Technology

Quarter-long Exchange. *Exchange major: Computer Science*

Sept. - Dec. 2021

## RESEARCH INTEREST

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*Research areas: Natural Language Processing, Reinforcement Learning, Programming Languages*

My research focuses on knowledge agents with LLMs, aiming to enable domain-independent approaches for in-depth exploration and synthesis, of information from diverse knowledge sources, including structured, unstructured, and hybrid sources. Some of my recent work includes reinforcement learning approaches ([SCRIBES](#), ICLR 2026; work done at Meta), agentic frameworks ([DataSTORM](#), pre-print; [SPINACH](#), EMNLP 2024), and programming-language interfaces for LLMs ([SUQL](#), NAACL 2024) to tackle this problem.

## SELECTED RESEARCH EXPERIENCE

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**AI Research Intern**

Jun. - Sept. 2025

Meta Reality Labs (in collaboration with FAIR at Meta)

**Research Assistant**

Jun. 2022 - Present

Stanford Open Virtual Assistant Lab & Stanford NLP Group

## PUBLICATIONS

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**DataSTORM: Deep Research on Large-Scale Databases using Exploratory Data Analysis and Data Storytelling**

**Shicheng Liu**, Yucheng Jiang, Sajid Farook, Camila Nicollier Sanchez, David Fernando Castro Pena, Monica S. Lam

*Pre-print, under review*

TL;DR: This paper introduces DataSTORM, an LLM-based agentic system that autonomously conducts research across large-scale structured databases and web sources, grounded in principles from Exploratory Data Analysis and Data Storytelling. We show that DataSTORM achieves state-of-the-art performance on existing benchmarks as well as a newly introduced real-world benchmark, outperforming systems like ChatGPT Deep Research in both automated metrics and human evaluations.

## SCRIBES: Web-Scale Script-Based Semi-Structured Data Extraction with Reinforcement Learning

**Shicheng Liu**, Kai Sun, Lisheng Fu, Xilun Chen, Xinyuan Zhang, Zhaojiang Lin, Rulin Shao, Yue Liu, Anuj Kumar, Wen-tau Yih, Xin Luna Dong

*Accepted to The Fourteenth International Conference on Learning Representations (ICLR 2026)*

TL;DR: SCRIBES is a reinforcement learning framework for large-scale extraction of semi-structured web data (tables, lists, infoboxes). Instead of running costly per-page LLM inference, it generates reusable scripts for groups of structurally similar pages, using layout similarity as a reward signal. Trained on both high-quality human annotations and synthetic CommonCrawl annotations, SCRIBES outperforms baselines by 13% in script quality and improves GPT-4o’s downstream QA accuracy by over 4%, making web information extraction more scalable and efficient, with more potential for complex QAs and model pre-training.

## SPINACH: SPARQL-Based Information Navigation for Challenging Real-World Questions

**Shicheng Liu\***, Sina J. Semnani\*, Harold Triedman, Jialiang Xu, Isaac Dan Zhao, Monica S. Lam.

*Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*

TL;DR: Introduces a novel Knowledge-Base Question Answering (KBQA) dataset and agent. The SPINACH dataset introduces challenging, expert-annotated questions from Wikidata’s request query forum. The SPINACH agent, mimicking how human experts write SPARQL queries, outperforms previous models across multiple KBQA datasets. [The deployed SPINACH agent](#) and the [online chat interface](#) have since been actively used by the Wikidata community, generating 22,000+ conversations.

[\[code\]](#) [\[video\]](#) [\[blog\]](#)

## SUQL: Conversational Search over Structured and Unstructured Data with Large Language Models

**Shicheng Liu**, Jialiang Xu, Wesley Tjangnaka, Sina J. Semnani, Chen Jie Yu, Monica S. Lam.

*Findings of the North American Chapter of the Association for Computational Linguistics: NAACL 2024*

TL;DR: Introduces the first conversational agent capable of accessing both structured and unstructured data from large knowledge corpora using a new language called SUQL (Structured and Unstructured Query Language), which extends SQL with free-text capabilities based on retrievers and LLMs. SUQL compiler performs important optimizations to power hybrid queries. Experiments on HybridQA and user studies on Yelp show that a SUQL-based agent outperforms strong baselines

[\[code\]](#) [\[video\]](#)

## Fine-tuned LLMs Know More, Hallucinate Less with Few-Shot Sequence-to-Sequence Semantic Parsing over Wikidata

Silei Xu\*, **Shicheng Liu\***, Theo Culhane, Elizaveta Pertseva, Meng-Hsi Wu, Sina Semnani, Monica Lam.

*Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*

TL;DR: Introduces WikiWebQuestions, a KBQA benchmark for Wikidata converted from the popular WebQuestionSP dataset. It presents a few-shot semantic parser based on fine-tuned version of LLaMA for Wikidata, with modified SPARQL syntax to enhance accuracy. When paired with GPT-3, the system can provide useful answers to 96% of the questions in the dev set of WikiWebQuestions.

[\[code\]](#) [\[video\]](#) [\[blog\]](#)

## Coding Reliable LLM-based Integrated Task and Knowledge Agents with GenieWorksheets

Harshit Joshi, **Shicheng Liu**, James Chen, Robert Weigle, Monica S. Lam.

*of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*

TL;DR: Introduces a programmable framework for creating task and knowledge conversational agents that handle complex interactions. GenieWorksheets enables developers to program agent policies through

its declarative paradigm. The compiled agent is resilient to diverse user queries and helpful with knowledge sources. It outperforms GPT-4 in execution accuracy, dialogue act accuracy, and goal completion rate, with results validated through real user studies.

**SPAGHETTI: Open-Domain Question Answering from Heterogeneous Data Sources with Retrieval and Semantic Parsing**

Heidi C. Zhang, Sina J. Semnani, Farhad Ghassemi, Jialiang Xu, **Shicheng Liu**, Monica S. Lam.

*Findings of the Association for Computational Linguistics: ACL 2024*

TL;DR: Introduces SPAGHETTI: Semantic Parsing Augmented Generation for Hybrid English information from Text Tables and Infoboxes, a hybrid question-answering (QA) pipeline that utilizes information from heterogeneous knowledge sources, including knowledge base, text, tables, and infoboxes. This LLM-augmented approach achieves SOTA performance on the Compmix dataset, the most comprehensive heterogeneous open-domain QA dataset

**Automated Testing of Software that Uses Machine Learning APIs**

Chengcheng Wan, **Shicheng Liu**, Sophie Xie, Yifan Liu, Henry Hoffmann, Michael Maire, Shan Lu.

*Proceedings of the 44th International Conference on Software Engineering, 2022*

**Are Machine Learning Cloud APIs Used Correctly?**

Chengcheng Wan, **Shicheng Liu**, Henry Hoffmann, Michael Maire, Shan Lu.

*Proceedings of the 43th International Conference on Software Engineering, 2021*

**SELECTED HONORS & AWARDS**

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Teachings

**Top-5% of Stanford CS Course Assistants:** Stanford CS224V 2023

Grants & Fellowships

**2025-2026 IBM PhD Fellowship Recipient** 2026

**2024 Brown Institute Magic Grant** 2024-2025

- Leading a bi-coastal collaboration between members from Stanford CS, Stanford Big Local News, and Columbia Journalism on *DataTalk: All Documents and Data, All at Once, All Verified*.
- *Project Overview:* Investigative journalism often relies on the ability to mine diverse data sets, with both structured and unstructured forms. Building on the novel programming language SUQL, this project aims to develop trustworthy conversational agents for journalists to uncover insights from hybrid data sources using natural-language queries. Example of published article using our agent on [Atlanta Journal Constitution](#) and [Honolulu Civil Beat](#).

Academic Honors

**Graduated Summa Cum Laude**, The University of Chicago 2022

**Outstanding Undergraduate Researcher Award**, Honorable Mention, CRA 2022

- [CRA website notice](#)
- Featured on [UChicago CS News](#)

**Elected member of Phi Beta Kappa**, the University of Chicago (the Beta chapter of Illinois) 2021

**Enrico Fermi Scholar**, The University of Chicago 2021

**Robert Maynard Hutchins Scholar**, The University of Chicago 2020

**Dean's List**, The University of Chicago 2018-2019, 2019-2020, 2020-2021

## TEACHING EXPERIENCES

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### Stanford

**Course Assistant** for CS 224N Conversational Virtual Assistants with Deep Learning      Winter 2026

**Head Course Assistant** for CS 224V Conversational Virtual Assistants with Deep Learning      Fall 2023

### University of Chicago

**Teaching Assistant** for DATA 12000 Computer Science for Data Science      Spring 2022

**Teaching Assistant** for CMSC 27200 Theory of Algorithms      Winter 2022

**Grader** for CMSC 22100 Programming Languages      Spring 2021

**Teaching Assistant** for CMSC 15100 Introduction to Computer Science I      Winter 2021

## ACADEMIC REFERENCES

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**Prof. Monica S. Lam** (*Ph.D. advisor*)

lam@cs.stanford.edu

Kleiner Perkins, Mayfield, Sequoia Capital Professor of the School of Engineering , Department of Computer Science

Stanford University, Stanford, U.S.A.

**Dr. Xin Luna Dong**

lunadong@meta.com

Principal Scientist, Meta Reality Labs

*This CV is last updated on April 24, 2026*